

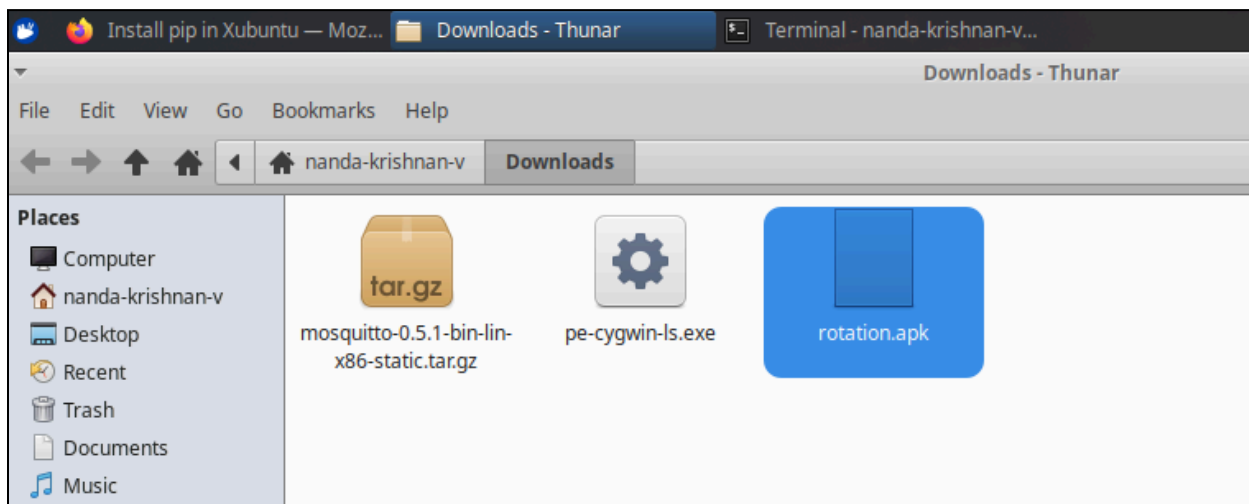
Model Lab Exam

Objective:

Download an APK file and perform a comprehensive static analysis using Androguard. Identify and document the following components:

- Permissions (including dangerous permissions)
- Activities
- Services
- Receivers
- Class names
- Extracted strings

Prepare a detailed technical report explaining all the identified elements and their potential security implications.



I have downloaded a random apk file from apkmirror and renamed the file to rotation.apk. Next, I have installed the androguard using `sudo apt install androguard`

```
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Downloads$ sudo apt install androguard
[sudo] password for nanda-krishnan-v:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
androguard is already the newest version (3.4.0~a1-10).
0 upgraded, 0 newly installed, 0 to remove and 73 not upgraded.
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Downloads$
```

Permissions (including dangerous permissions)

Command used: `a.get_permissions()`

```
In [14]: a.get_permissions()
...:
Out[14]:
['android.permission.PACKAGE_USAGE_STATS',
'android.permission.READ_PHONE_STATE',
'android.permission.ACCESS_AD SERVICES_TOPICS',
'android.permission.RECEIVE_BOOT_COMPLETED',
'android.permission.VIBRATE',
'android.permission.ACCESS_AD SERVICES_AD_ID',
'android.permission.WRITE_EXTERNAL_STORAGE',
'com.google.android.gms.permission.AD_ID',
'android.permission.WAKE_LOCK',
'android.permission.ACCESS_NETWORK_STATE',
'android.permission.QUERY_ALL_PACKAGES',
'android.permission.SYSTEM_ALERT_WINDOW',
'android.permission.FOREGROUND_SERVICE_SPECIAL_USE',
'android.permission.POST_NOTIFICATIONS',
'android.permission.GET_TASKS',
'android.permission.WRITE_SETTINGS',
'android.permission.READ_SETTINGS',
'android.permission.INTERNET',
'android.permission.USE_FULL_SCREEN_INTENT',
'com.pranavpandey.rotation.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION',
'com.android.vending.BILLING',
'com.pranavpandey.theme.permission.DYNAMIC_THEME',
'com.pranavpandey.theme.permission.READ_THEME',
'android.permission.ACCESS_AD SERVICES_ATTRIBUTION',
'android.permission.FOREGROUND_SERVICE']
```

Observations:

Dangerous and High-Risk Permissions

The most concerning findings are those permissions classified as Dangerous or granting significant system-level control:

- **android.permission.READ_PHONE_STATE (Dangerous):** This is a serious privacy concern. It allows the application to read the device's unique identifiers (like the IMEI), the user's phone number, and information about the current cellular network and ongoing calls. This capability is often used for device tracking and linking collected data to a specific user.

- **android.permission.WRITE_EXTERNAL_STORAGE (Dangerous):** Granting the ability to write to the shared external storage poses a risk of data corruption, deletion of user files, or, critically, the placement of malicious files or payloads that could be executed later.
- **android.permission.SYSTEM_ALERT_WINDOW (Dangerous):** This is a significant security red flag. It allows the app to draw over all other running applications. This capability is the foundation for highly effective attacks like tap-jacking or creating deceptive overlays (e.g., fake login screens for banking apps), making it a common feature in banking Trojans and ransomware.
- **android.permission.WRITE_SETTINGS (System Access):** This allows the app to modify system-level configurations, such as screen brightness, volume, and other core device behavior. Unauthorized modification of these settings could lead to significant user frustration or the disabling of important security or accessibility features.

Surveillance and Privacy Permissions

Several permissions indicate extensive user activity and application tracking, raising major privacy concerns:

- **android.permission.QUERY_ALL_PACKAGES (Sensitive):** This grants the ability to see a complete list of every single application installed on the device. This provides the attacker with a full profile of the user's interests, financial services, and security tools (e.g., antivirus).
- **android.permission.PACKAGE_USAGE_STATS and android.permission.GET_TASKS (Sensitive):** These allow the application to track which apps the user opens, how often, and for how long. This constitutes highly detailed surveillance of user habits and application usage history.
- **Advertising Permissions (AD_ID, ADSERVICES_TOPICS, etc.):** The presence of multiple Google Mobile Services (GMS) permissions related to advertising indicates that the app is deeply integrated with user profiling and tracking for monetization purposes.

Persistence and Resource Usage

Permissions used to maintain control and consume resources include:

- **android.permission.RECEIVE_BOOT_COMPLETED:** This is a classic method for achieving persistence. It ensures the application automatically starts and resumes its operations (malicious or otherwise) every time the device reboots.
- **android.permission.WAKE_LOCK:** This allows the app to prevent the CPU from sleeping, enabling background processes to run indefinitely. This can lead to excessive battery drain and ensures continuous, hidden operation for surveillance or communication with a Command and Control (C2) server.
- **android.permission.FOREGROUND_SERVICE:** While legitimate for some apps, this allows the service to run with elevated priority, often hidden from the user, making it difficult to terminate and ensuring that the app's functions remain active.

Activities

```
In [15]: a.get_activities()
...:
Out[15]:
['com.pranavpandey.rotation.activity.SetupActivity',
'com.android.billingclient.api.ProxyBillingActivity',
'com.pranavpandey.rotation.activity.HomeActivity',
'com.pranavpandey.android.dynamic.support.permission.activity.DynamicPermissionsActivity',
'com.pranavpandey.rotation.activity.ActionActivity',
'com.pranavpandey.rotation.activity.EditActivity',
'com.google.android.gms.common.api.GoogleApiActivity',
'com.pranavpandey.rotation.activity.ExtensionActivity',
'com.pranavpandey.android.dynamic.support.preview.activity.DynamicPreviewActivity',
'com.pranavpandey.rotation.activity.SplashActivity',
'com.google.android.gms.ads.OutOfContextTestingActivity',
'com.pranavpandey.rotation.activity.PreviewActivity',
'com.pranavpandey.rotation.activity.ShortcutsActivity',
'com.pranavpandey.android.dynamic.billing.activity.DynamicBillingActivity',
'com.google.android.gms.ads.NotificationHandlerActivity',
'com.pranavpandey.rotation.activity.WidgetActivity',
'com.pranavpandey.rotation.activity.PermissionsActivity',
'com.pranavpandey.rotation.activity.TutorialActivity',
'com.google.android.gms.ads.AdActivity',
'com.pranavpandey.rotation.activity.ThemeActivity',
'com.android.billingclient.api.ProxyBillingActivityV2']
```

The analysis of the Activities reveals the core functionality and entry points of the application, confirming the app's high level of complexity and integration with third-party services.

Core Functionality & Security Focus

- **Setup, Permissions, and Dynamic Activities:** The presence of **SetupActivity**, **DynamicPermissionsActivity**, and **PermissionsActivity** confirms the app manages sensitive runtime permissions (like those identified earlier, e.g., **SYSTEM_ALERT_WINDOW**). This highlights the importance of these screens, as they are the gates to the app's invasive functionality.
- **Monetization Activities:** Multiple **Billing Activities** (**ProxyBillingActivity**, **DynamicBillingActivity**) confirm the use of Google Play In-App Billing. This requires protection against **billing bypass fraud** and could pose a risk if the payment logic is not securely implemented.
- **Advertising Integration:** Activities like **AdActivity**, **NotificationHandlerActivity**, and others associated with Google Mobile Services (GMS) confirm the app's heavy reliance on **AdMob**. This directly links to the numerous advertising permissions and suggests the app is primarily focused on **tracking and user profiling** for targeted ad delivery.

Security Implications

The main risks stem from:

1. **Unauthorized Access:** If sensitive activities (like those handling setup or permissions) are **exported** (accessible by other apps), they could allow a malicious application to **bypass security checks** and launch privileged functions directly.
2. **Privacy/Adware Concern:** The combination of dedicated permission-handling screens and deep advertising integration strongly suggests the app is aggressive in both **data collection** and **user tracking**, aligning with potential adware characteristics identified in the Permissions section.

Services

```
In [16]: a.get_services()
...:
Out[16]:
['com.pranavpandey.rotation.service.RotationTileOrientation',
'com.google.android.datatransport.runtime.backends.TransportBackendDiscovery',
'com.google.android.datatransport.runtime.scheduling.jobscheduling.JobInfoSchedulerService',
'com.pranavpandey.rotation.service.RotationTileService',
'androidx.work.impl.background.systemalarm.SystemAlarmService',
'androidx.work.impl.background.systemjob.SystemJobService',
'androidx.work.impl.foreground.SystemForegroundService',
'com.pranavpandey.rotation.service.RotationService',
'com.google.android.gms.ads.AdService',
'androidx.room.MultiInstanceInvalidationService']
```

The extracted services fall into three main functional categories:

1. **Core Application Services:**
 - **com.pranavpandey.rotation.service.RotationService and RotationTileService:** These services are central to the app's functionality (likely related to managing screen rotation and Quick Settings Tiles). Since they run constantly in the background, they are the primary targets for any system-level control the app requires (e.g., executing the changes enabled by the WRITE_SETTINGS permission).
 - **RotationTileOrientation:** A helper service likely managing the state of the Quick Settings tile.
2. **Background Task Management and Persistence Services:**
 - **androidx.work.impl.background.SystemAlarmService, SystemJobService, and SystemForegroundService:** These indicate the app uses Android's **WorkManager** library. WorkManager is a legitimate tool for deferred and guaranteed background execution. However, its use guarantees that the app's tasks will **persist** even after the app is closed, which is a key characteristic of malware trying to maintain a foothold.

- The presence of **SystemForegroundService** directly correlates with the **FOREGROUND_SERVICE** permission, meaning the app runs a visible (but often non-interactive) persistent notification to keep the background tasks alive and exempt from battery optimization.
3. **Third-Party and Data Services:**
- **com.google.android.gms.ads.AdService:** Confirms the background component of the Google Ads integration. This service is responsible for handling ad delivery, fetching targeting information, and running ad-related tracking processes **silently in the background**.
 - **Services related to com.google.android.datatransport (e.g., TransportBackendDiscovery, JobInfoSchedulerService):** These are used for reliable and efficient background data transfer, often for analytics, logging, or reporting user activity back to a server.

Security Implications

The Services are the engine for the app's high-risk behavior:

1. **Guaranteed Persistence:** The service architecture, combined with prior permissions (like **RECEIVE_BOOT_COMPLETED**), ensures the app maintains an **uninterruptible presence** on the device, a hallmark of malicious or highly aggressive applications.
2. **Covert Data Exfiltration:** These services provide the necessary channel and time for the application to **covertly collect and send** the sensitive data authorized by permissions (**READ_PHONE_STATE**, **PACKAGE_USAGE_STATS**) back to external servers, often linked to the monetization or advertising goals.
3. **Export Risk:** If any custom service is **exported** in the Manifest, it creates a potential **escalation vulnerability**, allowing other apps to trigger powerful system-altering operations.

Receivers

```
In [17]: a.get_receivers()
...:
Out[17]:
['com.pranavpandey.rotation.receiver.ThemeReceiver',
'com.pranavpandey.rotation.receiver.ActionReceiver',
'com.pranavpandey.android.dynamic.support.theme.receiver.DynamicColorsReceiver',
'com.pranavpandey.android.dynamic.engine.receiver.DynamicStateReceiver',
'androidx.work.impl.utils.ForceStopRunnable$BroadcastReceiver',
'androidx.work.impl.background.systemalarm.ConstraintProxy$NetworkStateProxy',
'androidx.work.impl.background.systemalarm.RescheduleReceiver',
'androidx.work.impl.diagnostics.DiagnosticsReceiver',
'androidx.work.impl.background.systemalarm.ConstraintProxyUpdateReceiver',
'com.pranavpandey.rotation.receiver.UpdateReceiver',
'com.pranavpandey.rotation.provider.TogglesWidgetProvider',
'androidx.work.impl.background.systemalarm.ConstraintProxy$BatteryChargingProxy',
'androidx.work.impl.background.systemalarm.ConstraintProxy$BatteryNotLowProxy',
'com.pranavpandey.rotation.provider.ServiceWidgetProvider',
'androidx.work.impl.background.systemalarm.ConstraintProxy$StorageNotLowProxy',
'androidx.profileinstaller.ProfileInstallReceiver',
'com.google.android.datatransport.runtime.scheduling.jobscheduling.AlarmManagerSchedulerBroadcastReceiver',
'com.pranavpandey.rotation.receiver.BootReceiver']
```

The Broadcast Receivers confirm critical mechanisms for persistence, system monitoring, and guaranteed background execution.

Observations and Security Implications

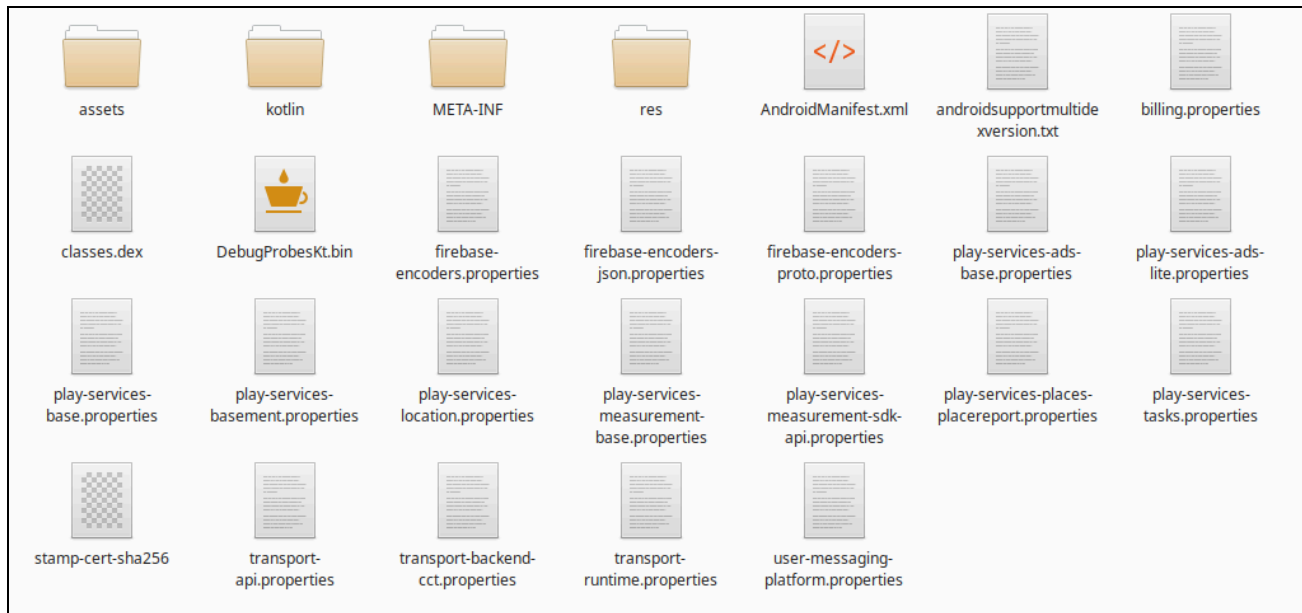
- **Persistence (Primary Concern):**
 - **com.pranavpandey.rotation.receiver.BootReceiver:** Directly linked to the `RECEIVE_BOOT_COMPLETED` permission.
 - **Security Implication:** Ensures the application achieves **maximum persistence**, automatically restarting its invasive services immediately after the device boots up.
- **Resilience and Recovery:**
 - **WorkManager Receivers (ForceStopRunnable\$BroadcastReceiver, RescheduleReceiver, etc.):** Components designed to monitor the system state.
 - **Security Implication:** Allows the app to detect if it has been force-stopped by the user and **automatically reschedule/restart** its persistent background tasks, making it highly difficult to terminate.
- **Covert System Monitoring:**
 - **Constraint Proxy Receivers (NetworkStateProxy, BatteryChargingProxy, etc.):** Enable the app to trigger tasks based on specific system conditions (e.g., when the network is available or the battery is full).
 - **Security Implication:** Grants the app **fine-grained, covert control** to execute high-risk operations (like data exfiltration) only when conditions are optimal and least likely to be noticed.
- **Potential Export Vulnerability:**

- ## Class names

JADX (Java Decompiler for Android Dex) is a popular and powerful open-source tool used primarily for **reverse engineering** and **static analysis** of Android applications.

Setting up the path:

Next, I have used the command - `jadx -d ~/Desktop/rotation_output/rotation.apk` to decompile the apk into the new file named `rotation_output`. In this file we could get all the data in the apk including the strings and so on.



This is the file structure that is produced after executing the command.

```
rotation_output/
├── sources/                                ← Decompiled Java source files (.java)
│   ├── com/
│   │   └── example/
│   │       └── rotation/
│   │           ├── MainActivity.java
│   │           └── utils/
│   │               └── ScreenHelper.java
├── resources/                             ← Decompiled resources (drawables, XMLs)
├── AndroidManifest.xml                    ← Decompiled Android manifest (not original)
└── jadx-project.json                     ← Metadata for JADX
```

This is the structure of the Directory.

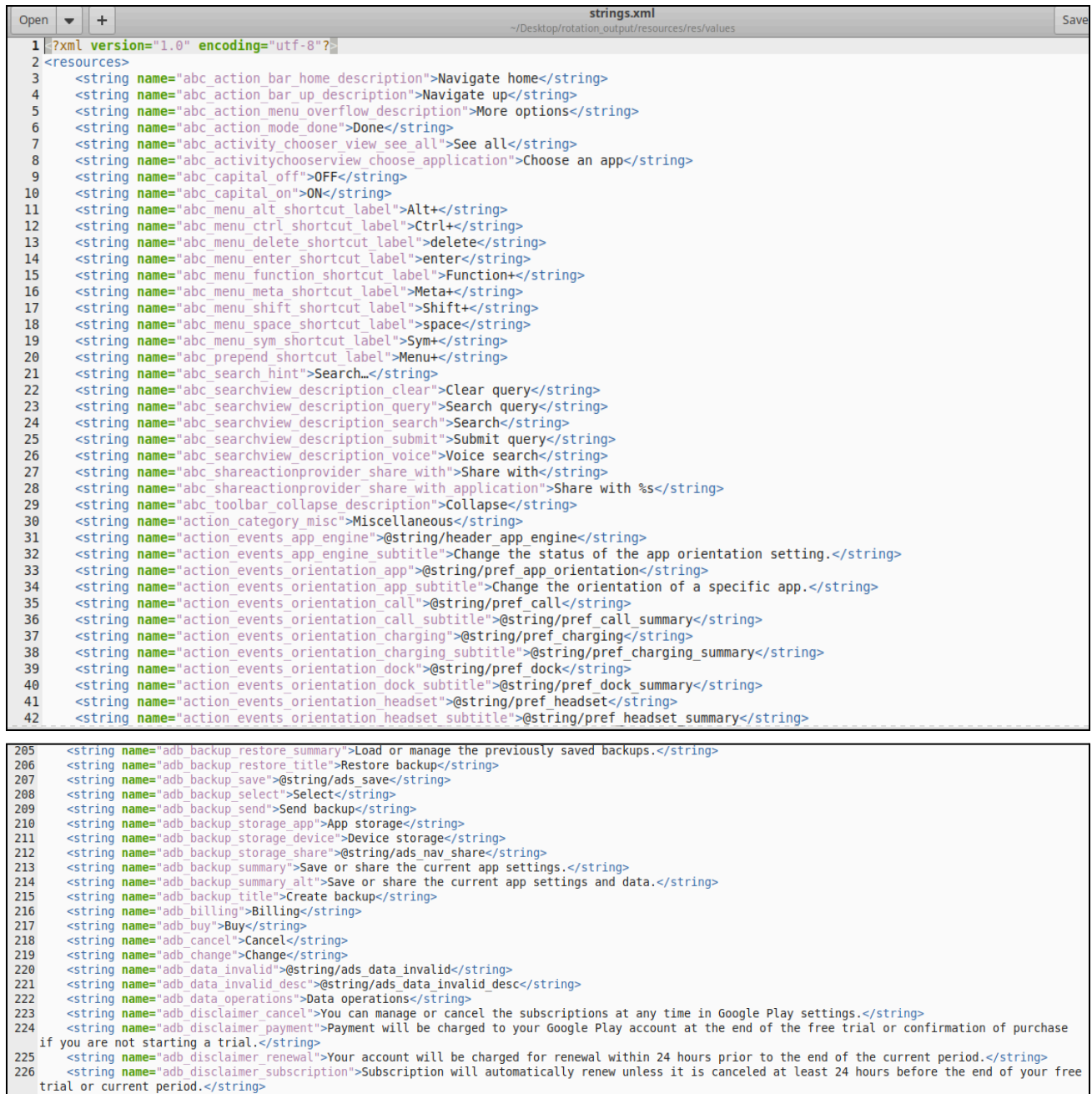
Extracted strings

```
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop$ sudo apt install gedit -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gedit is already the newest version (46.2-2).
0 upgraded, 0 newly installed, 0 to remove and 75 not upgraded.
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop$
```

For extracting the string I have installed another tool called gedit. Gedit is a lightweight, free, and open-source text editor primarily designed for the GNOME desktop environment, known for its elegant simplicity and user-friendly Graphical User Interface (GUI).

This is because while extracting the class we could get the strings also, they are placed inside the rotation_output file, which is xml format. We need to open it in the text editor.

These are the observed screenshot of the strings.xml



```
1  ?xml version="1.0" encoding="utf-8"?
2  <resources>
3      <string name="abc_action_bar_home_description">Navigate home</string>
4      <string name="abc_action_bar_up_description">Navigate up</string>
5      <string name="abc_action_menu_overflow_description">More options</string>
6      <string name="abc_action_mode_done">Done</string>
7      <string name="abc_activity_chooser_view_see_all">See all</string>
8      <string name="abc_activitychooserview_choose_application">Choose an app</string>
9      <string name="abc_capital_off">OFF</string>
10     <string name="abc_capital_on">ON</string>
11     <string name="abc_menu_alt_shortcut_label">Alt+</string>
12     <string name="abc_menu_ctrl_shortcut_label">Ctrl+</string>
13     <string name="abc_menu_delete_shortcut_label">delete</string>
14     <string name="abc_menu_enter_shortcut_label">enter</string>
15     <string name="abc_menu_function_shortcut_label">Function+</string>
16     <string name="abc_menu_meta_shortcut_label">Meta+</string>
17     <string name="abc_menu_shift_shortcut_label">Shift+</string>
18     <string name="abc_menu_space_shortcut_label">space</string>
19     <string name="abc_menu_sym_shortcut_label">Sym+</string>
20     <string name="abc_prepend_shortcut_label">Menu+</string>
21     <string name="abc_search_hint">Search...</string>
22     <string name="abc_searchview_description_clear">Clear query</string>
23     <string name="abc_searchview_description_query">Search query</string>
24     <string name="abc_searchview_description_search">Search</string>
25     <string name="abc_searchview_description_submit">Submit query</string>
26     <string name="abc_searchview_description_voice">Voice search</string>
27     <string name="abc_shareactionprovider_share_with">Share with</string>
28     <string name="abc_shareactionprovider_share_with_application">Share with %s</string>
29     <string name="abc_toolbar_collapse_description">Collapse</string>
30     <string name="action_category_misc">Miscellaneous</string>
31     <string name="action_events_app_engine">@string/header_app_engine</string>
32     <string name="action_events_app_engine_subtitle">Change the status of the app orientation setting.</string>
33     <string name="action_events_orientation_app">@string/pref_app_orientation</string>
34     <string name="action_events_orientation_app_subtitle">Change the orientation of a specific app.</string>
35     <string name="action_events_orientation_call">@string/pref_call</string>
36     <string name="action_events_orientation_call_subtitle">@string/pref_call_summary</string>
37     <string name="action_events_orientation_charging">@string/pref_charging</string>
38     <string name="action_events_orientation_charging_subtitle">@string/pref_charging_summary</string>
39     <string name="action_events_orientation_dock">@string/pref_dock</string>
40     <string name="action_events_orientation_dock_subtitle">@string/pref_dock_summary</string>
41     <string name="action_events_orientation_headset">@string/pref_headset</string>
42     <string name="action_events_orientation_headset_subtitle">@string/pref_headset_summary</string>
...
205 <string name="adb_backup_restore_summary">Load or manage the previously saved backups.</string>
206 <string name="adb_backup_restore_title">Restore backup</string>
207 <string name="adb_backup_save">@string/ads_save</string>
208 <string name="adb_backup_select">Select</string>
209 <string name="adb_backup_send">Send backup</string>
210 <string name="adb_backup_storage_app">App storage</string>
211 <string name="adb_backup_storage_device">Device storage</string>
212 <string name="adb_backup_storage_share">@string/ads_nav_share</string>
213 <string name="adb_backup_summary">Save or share the current app settings.</string>
214 <string name="adb_backup_summary_alt">Save or share the current app settings and data.</string>
215 <string name="adb_backup_title">Create backup</string>
216 <string name="adb_billing">Billing</string>
217 <string name="adb_buy">Buy</string>
218 <string name="adb_cancel">Cancel</string>
219 <string name="adb_change">Change</string>
220 <string name="adb_data_invalid">@string/ads_data_invalid</string>
221 <string name="adb_data_invalid_desc">@string/ads_data_invalid_desc</string>
222 <string name="adb_data_operations">Data operations</string>
223 <string name="adb_disclaimer_cancel">You can manage or cancel the subscriptions at any time in Google Play settings.</string>
224 <string name="adb_disclaimer_payment">Payment will be charged to your Google Play account at the end of the free trial or confirmation of purchase if you are not starting a trial.</string>
225 <string name="adb_disclaimer_renewal">Your account will be charged for renewal within 24 hours prior to the end of the current period.</string>
226 <string name="adb_disclaimer_subscription">Subscription will automatically renew unless it is canceled at least 24 hours before the end of your free trial or current period.</string>
```

```

292 <string name="adb_validity_hour">%1$d hour</string>
293 <string name="adb_validity_hours">%1$d hours</string>
294 <string name="adb_validity_month">%1$d month</string>
295 <string name="adb_validity_months">%1$d months</string>
296 <string name="adb_validity_week">%1$d week</string>
297 <string name="adb_validity_weeks">%1$d weeks</string>
298 <string name="adb_validity_year">%1$d year</string>
299 <string name="adb_validity_years">%1$d years</string>
300 <string name="adk_app_key">Rotation Key</string>
301 <string name="adk_app_key_buy">@string/adk_buy</string>
302 <string name="adk_app_key_buy_desc">Buy %1$s from %2$s to unlock the following features:</string>
303 <string name="adk_app_key_buy_desc_gen">Buy %1$s from %2$s to unlock all the features.</string>
304 <string name="adk_app_key_installed">Installed</string>
305 <string name="adk_app_key_installed_desc">Thanks for purchasing the %1$s. Now, the following features are available for use:</string>
306 <string name="adk_app_key_installed_desc_gen">Thanks for purchasing the %1$s. Now, all the features are available for use.</string>
307 <string name="adk_app_key_not_installed">Not installed</string>
308 <string name="adk_app_key_not_installed_desc">%1$s is not installed. Purchase it from %2$s to unlock the following features:</string>
309 <string name="adk_app_key_not_installed_desc_gen">%1$s is not installed. Purchase it from %2$s to unlock all the features.</string>
310 <string name="adk_app_key_removed_desc">%1$s has been removed. Please install it again to use the following features:</string>
311 <string name="adk_app_key_removed_desc_gen">%1$s has been removed. Please install it again to use all the features.</string>
312 <string name="adk_app_key_short">Key</string>
313 <string name="adk_buy">Buy</string>
314 <string name="adk_purchase">Purchase</string>
315 <string name="adr_contribute">Contribute</string>
316 <string name="adr_feedback">Feedback</string>
317 <string name="adr_later">Later</string>
318 <string name="adr_rate">Rate</string>
319 <string name="adr_rate_feedback_desc">Please rate to support the development or provide feedback to make things even better.</string>
320 <string name="adr_skip">Skip</string>
321 <string name="ads_about">About</string>
322 <string name="ads_accept">Accept</string>
323 <string name="ads_actions">Actions</string>
324 <string name="ads_advanced">Advanced</string>
325 <string name="ads_app">App</string>
326 <string name="ads_author">Author</string>
327 <string name="ads_cancel">@android:string/cancel</string>
328 <string name="ads_clear">Clear</string>
329 <string name="ads_color">Color</string>
330 <string name="ads_configure">Configure</string>
331 <string name="ads_contributors">Contributors</string>
332 <string name="ads_contributors_desc">Huge thanks for their great support.</string>

```

```

1272 <string name="tutorial_finish_desc">I am glad you have decided to try Rotation. I hope you will enjoy all the features of this app. Run setup to
1273 configure the important features with ease. Please rate and feel free to write any feedback.</string>
1274 <string name="tutorial_finish_directions">Get more info via Rotation &gt; Support &gt; Help.</string>
1275 <string name="tutorial_finish_subtitle">Run setup to configure</string>
1276 <string name="tutorial_global_orientation">@string/mode_global</string>
1277 <string name="tutorial_global_orientation_desc">@string/mode_global_desc</string>
1278 <string name="tutorial_global_orientation_directions">It can be selected via Rotation &gt; Home &gt; Orientation.</string>
1279 <string name="tutorial_global_orientation_subtitle">Default orientation</string>
1280 <string name="tutorial_key">@string/adk_app_key</string>
1281 <string name="tutorial_key_desc">Buy to support the development and unlock features like rotate on demand, data operations and many more. Automate
most of the actions via Tasker / Locale plugin.</string>
1282 <string name="tutorial_key_desc_available">Paid features like rotate on demand, data operations, etc. are now available for use. Automate most of
the actions via Tasker / Locale plugin.</string>
1283 <string name="tutorial_key_directions">Key status can be checked via Rotation &gt; About &gt; Info &gt; Rotation Key.</string>
1284 <string name="tutorial_key_subtitle">On demand, extension and more</string>
1285 <string name="tutorial_label">Rotation tutorial</string>
1286 <string name="tutorial_more">Settings</string>
1287 <string name="tutorial_more_desc">Various settings, notification toggles, shortcuts and widgets can be customized according to the requirements. A
bunch of themes and unlimited colors to provide a style for every choice.</string>
1288 <string name="tutorial_more_subtitle">They can be accessed via Rotation &gt; Settings &gt; App, Notification or Widgets. Widgets and shortcuts can
be added via launcher (home screen).</string>
1289 <string name="tutorial_test_release">App, notification and widgets</string>
1290 <string name="tutorial_test_release_accept">Rotation reboot</string>
1291 <string name="tutorial_test_release_desc">@string/ads_yes_in</string>
1292 <string name="tutorial_test_release_desc">A complete reboot to make things stable and even better!\n\nThis version is completely rebuilt from
scratch to provide more control and better stability. Your previous settings (below version 8.0.0) will not work from this release and it is not
translated in all the languages. Rotation is in active development and I will do my best to add more languages as soon as possible.\n\nPlease join
official community to provide feedback, translate or to submit bug reports.</string>
1293 <string name="tutorial_test_release_reject">@string/ads_not_now</string>
1294 <string name="tutorial_welcome">@string/app_name</string>
1295 <string name="tutorial_welcome_desc">Welcome to Rotation, the most intuitive and customizable orientation manager for Android. Please read this
quick tutorial to get familiar with most of the features that Rotation provides.</string>
1296 <string name="tutorial_welcome_subtitle">@string/app_subtitle</string>
1297 <string name="v7_preference_off">OFF</string>
1298 <string name="v7_preference_on">ON</string>
1299 <string name="widget_service">Service</string>
1300 <string name="widget_service_long">Service widget</string>
1301 <string name="widget_toggles">Toggles</string>
1302 <string name="widget_toggles_edit">Widget toggles</string>
1303 <string name="widget_toggles_long">Toggles widget</string>
1303 </resources>

```

Observations:

What does the app do?

- It's a powerful app for managing your phone's screen **rotation and orientation**.
- It works by tapping into deep system features, using special **Activities, Services** (like a Quick Settings tile), and **Receivers** to handle events like a phone starting up.
- It needs many permissions to work, including those to **write system settings** and draw over other apps (SYSTEM_ALERT_WINDOW).

How the App Makes Money

- The app is a freemium model.
- Users can buy an "adk app key" or "Rotation Key" to unlock all features.
- It uses Google Play Billing and supports subscriptions with free trials and auto-renewal.
- It includes ads (Google AdService).

Technical Information

- The analysis was performed by running the Android app's file (rotation.apk) through a decompiler tool called JADX.
- The core package name is com.pranavpandey.rotation.

Deep System Integration

- The app requires highly sensitive system permissions such as WRITE_SETTINGS and SYSTEM_ALERT_WINDOW to function.
- It registers a BootReceiver to start on device boot.
- It implements a core RotationService and a RotationTileService (Quick Settings tile).
- Advanced features include integration with automation tools like Tasker / Locale plugin.